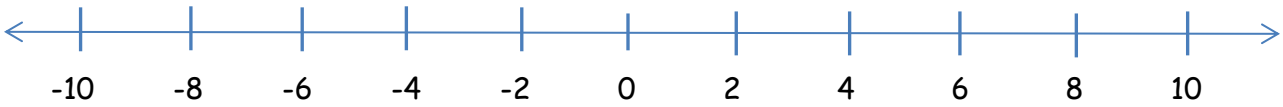
Algebra 1 Name_____

Why do we need to flip the inequality sign??

Use the number line to help you answer the questions in the table below:



First Inequality	True or False?	Action	New Inequality	True or False?
1 < 4	If true proceed	Add 3 to both sides		
-4 < 6	If true proceed	Subtract 3 from both sides		
-1 < 5	If true proceed	multiply both sides by 2		
-3 < 3	If true proceed	multiply both sides by -2		
-2 < 10	If true proceed	divide both sides by 2		
2 < 8	If true proceed	divide both sides by -2		

Use your work from the table above to answer the following questions:

1) Which **specific actions** caused the first inequality to become a false inequality?

2) Without changing the action, what could be done to the new inequality to make it true?

3) Summarize what you learned here.

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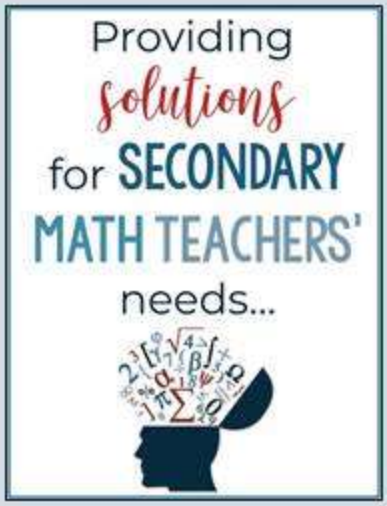

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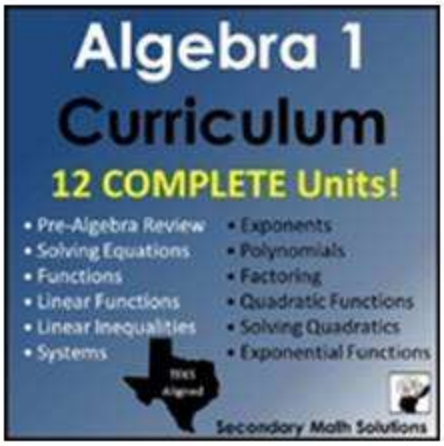

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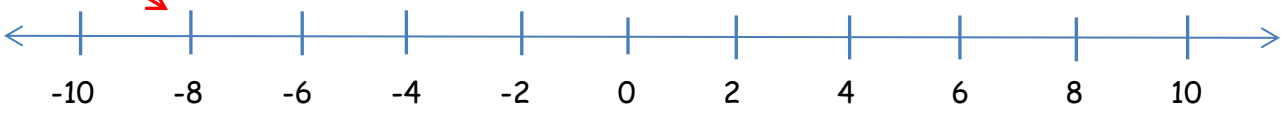
Algebra 1 Name_____Key_____

Why do we need to flip the inequality sign??

You might need to remind students that as we move to the right on the number line, the numbers get larger

Use the number line to help you answer the questions in the table below:



First Inequality	True or False?	Action	New Inequality	True or False?
1 < 4	true If true proceed	Add 3 to both sides	4 < 7	true
-4 < 6	true If true proceed	Subtract 3 from both sides	-7 < 3	true
-1 < 5	true If true proceed	multiply both sides by 2	-2 < 10	true
-3 < 3	true If true proceed	multiply both sides by -2	6 < -6	false
-2 < 10	true If true proceed	divide both sides by 2	-1 < 5	true
2 < 8	true If true proceed	divide both sides by -2	-1 < -4	false

You might have them circle or highlight the two that "didn't work"

Use your work from the table above to answer the following questions:

1) Which **specific actions** caused the first inequality to become a false inequality?

Multiplying or dividing by a negative number

2) Without changing the action, what could be done to the new inequality to make it true?

Flip the inequality sign

3) Summarize what you learned here.

If an inequality is multiplied or divided on both sides by a NEGATIVE number, then the inequality sign needs to be flipped in order to maintain the truth of the inequality.

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